**Examples** : Phenacite - $Be_2SiO_4$ ($Be^{2+}$ ions are tetrahedrally surrounded by $O^{2-}$ ions), Olivine - $(Fe/Mg)_2SiO_4$ ( $Fe^{2+}$ and $Mg^{2+}$ cations are octahedrally surrounded by $O^{2-}$ ions),

**Pyro silicate (or) Soro silicates):** Silicates which contain $[Si_2O_7]^{6-}$ ions are called pyro silicates (or) Soro silicates. They are formed by joining two $[SiO_4]^{4-}$ tetrahedral units by sharing one oxygen atom at one corner.(one oxygen is removed while joining). Example : Thortveitite - $Sc_2Si_2O_7$

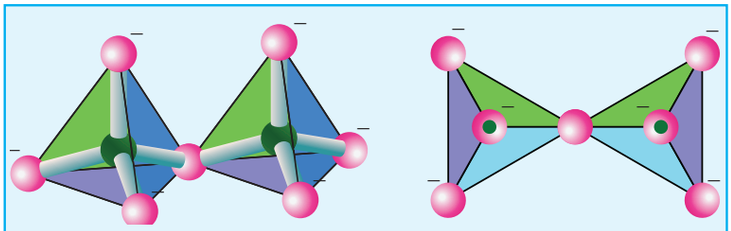


Figure 2.12 Structure of Pyro silicate

## Cyclic silicates (or Ring silicates)

Silicates which contain $(SiO_3)_n^{2n-}$ ions which are formed by linking three or more tetrahedral $SiO_4^{4-}$ units cyclically are called cyclic silicates. Each silicate unit shares two of its oxygen atoms with other units.

**Example:** Beryl $[Be_3Al_2(SiO_3)_6]$ (an aluminosilicate with each aluminium is surrounded by 6 oxygen atoms octahedrally)

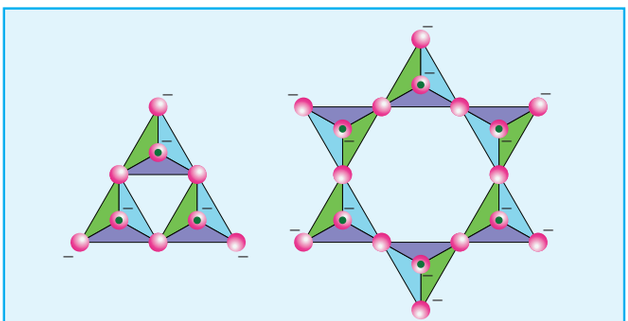


Figure 2.13 Structure of Cyclic silicates

## Inosilicates :

Silicates which contain 'n' number of silicate units linked by sharing two or more oxygen atoms are called inosilicates. They are further classified as chain silicates and double chain silicates.

**Chain silicates (or pyroxenes):** These silicates contain $[(SiO_3)_n]^{2n-}$ ions formed by linking 'n' number of tetrahedral $[SiO_4]^{4-}$ units linearly. Each silicate unit shares two of its oxygen atoms with other units.

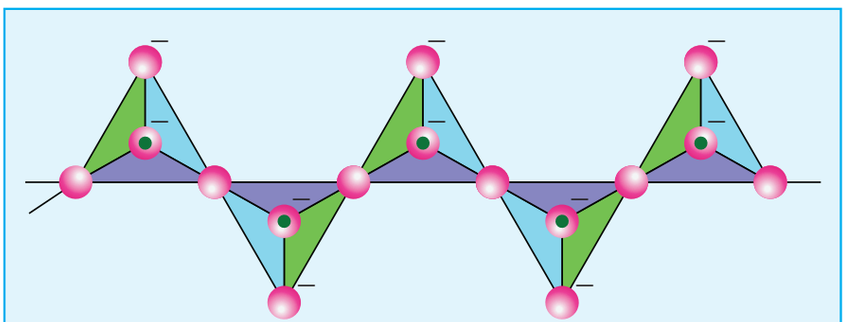


Figure 2.14 Structure of Chain silicates

**Example:** Spodumene - $LiAl(SiO_3)_2$.

**Double chain silicates (or amphiboles):** These silicates contains $[Si_4O_{11}]_n^{6n-}$ ions. In these silicates there are two different types of tetrahedra : (i) Those sharing 3 vertices (ii) those sharing only 2 vertices.



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